The Interaction

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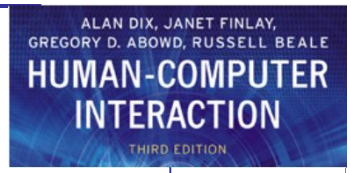
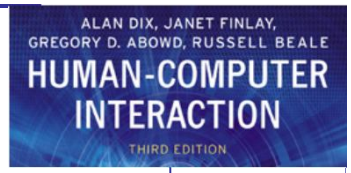


- What is interaction?

  communication

  user ⟷ system

- interaction models
  - translations between user and system
- ergonomics
  - physical characteristics of interaction
- interaction styles
  - the nature of user/system dialog
- context
  - social, organizational, motivational



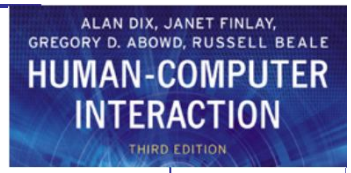
Models of Interaction

terms of interaction

Norman model

interaction framework

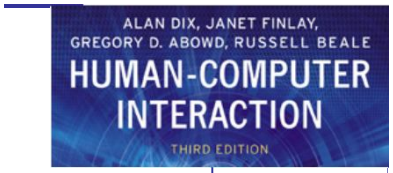
Some Terms of Interaction



- The purpose of an interactive system is to aid a user in accomplishing goals from some application domain.

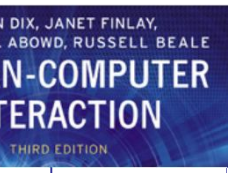
- Domain – the area of work under study
  e.g. graphic design
- goal – what you want to achieve
  e.g. create a solid red triangle
- task – how you go about doing it
  – ultimately in terms of operations or actions
  e.g. … select fill tool, click over triangle

Donald Norman's model

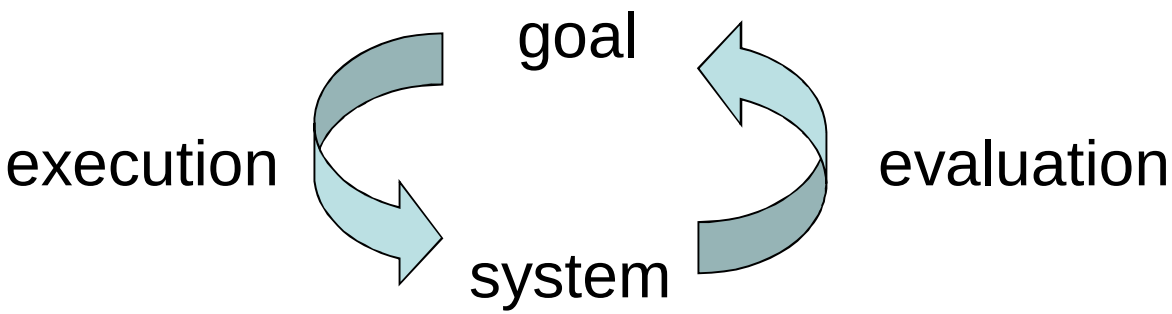


- Perhaps the most influential in Human–Computer Interaction, possibly because of its closeness to our intuitive understanding of the interaction between human user and computer
- The user formulates a plan of action, which is then executed at the computer interface.
- When the plan, or part of the plan, has been executed, the user observes the computer interface to evaluate the result of the executed plan, and to determine further actions.
- The interactive cycle can be divided into two major phases: execution and evaluation.

Donald Norman's model - Seven stages



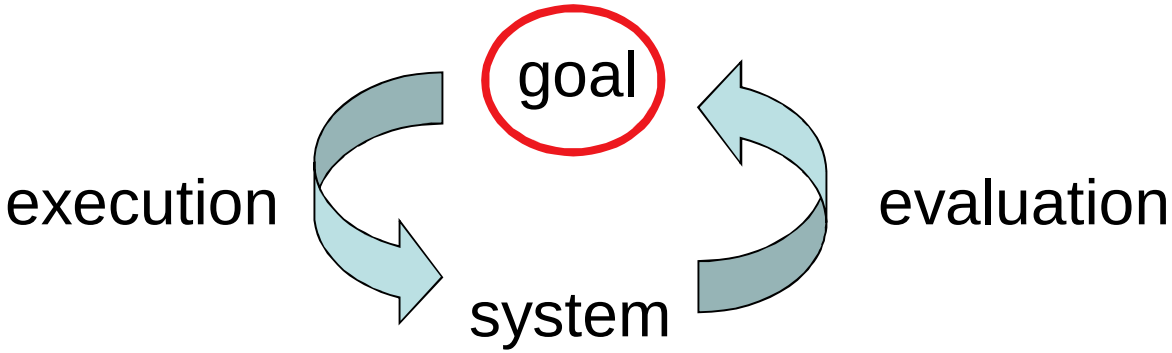
- execution/evaluation loop



  - user establishes the goal
  - formulates intention
  - specifies actions at interface
  - executes action
  - perceives system state
  - interprets system state
  - evaluates system state with respect to goal

- Norman's model concentrates on user's view of the interface

execution/evaluation loop



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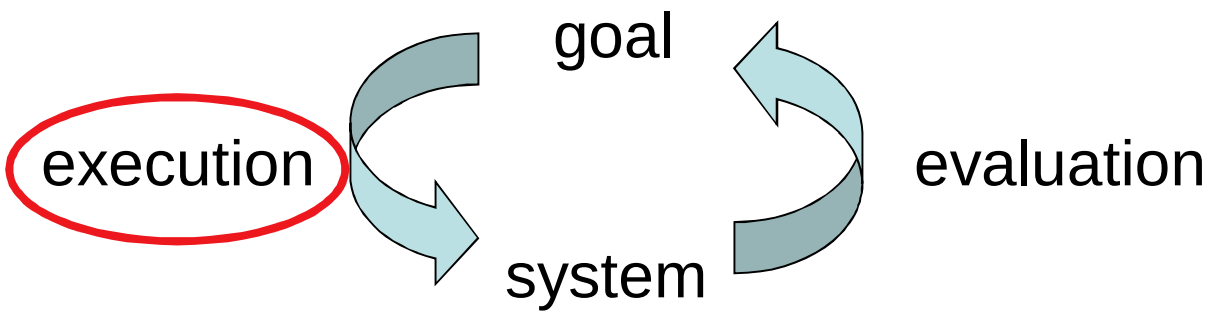

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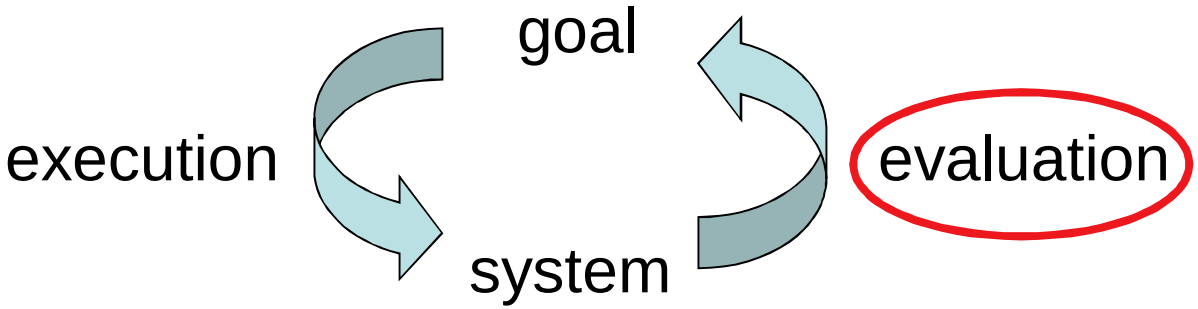

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Using Norman's Model

Some systems are harder to use than others

Gulf of Execution

user's formulation of actions

≠ actions allowed by the system

Gulf of Evaluation

user's expectation of changed system state

≠ actual presentation of this state

Human Error - Slips and Mistakes

slip

- understand system and goal
- correct formulation of action
- incorrect action

mistake

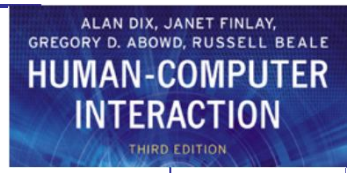
- may not even have right goal!

Fixing things?

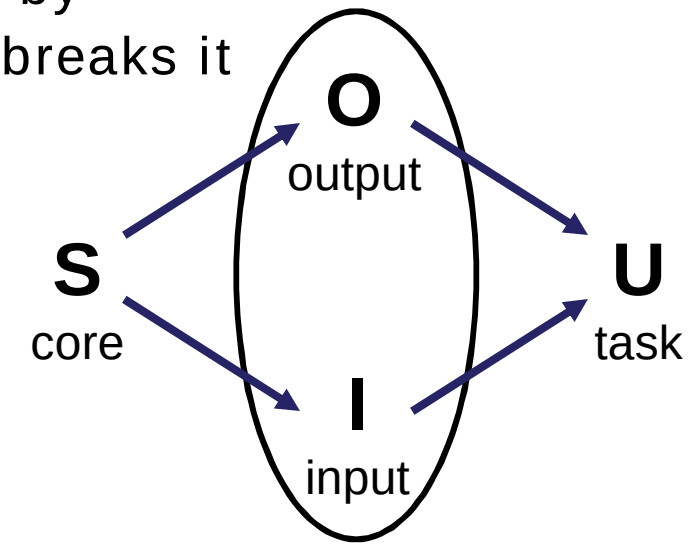
slip – better interface design

mistake – better understanding of system

Abowd and Beale Framework



- Extension of Norman…
- The interaction framework attempts a more realistic description of interaction by including the system explicitly, and breaks it into four main components:
  - user
  - input
  - system
  - output



- each has its own unique language

  interaction ⇒ translation between languages

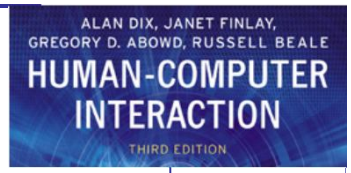
- problems in interaction = problems in translation

Using Abowd & Beale's Model

user intentions
→ translated into actions at the interface
→ translated into alterations of system state
→ reflected in the output display
→ interpreted by the user

general framework for understanding interaction

- not restricted to electronic computer systems
- identifies all major components involved in interaction
- allows comparative assessment of systems
- an abstraction



Ergonomics

physical aspects of interfaces

industrial interfaces

Ergonomics

- Study of the physical characteristics of interaction
- Also known as human factors – but this can also be used to mean much of HCI!
- Ergonomics good at defining standards and guidelines for constraining the way we design certain aspects of systems

Ergonomics - Examples

- arrangement of controls and displays
  - e.g. controls grouped according to function or frequency of use, or sequentially
- surrounding environment
  - e.g. seating arrangements adaptable to cope with all sizes of user
- health issues
  - e.g. physical position, environmental conditions (temperature, humidity), lighting, noise,
- use of colour
  - e.g. use of red for warning, green for okay, awareness of colour-blindness etc.

Industrial Interfaces

Office interface vs. industrial interface?

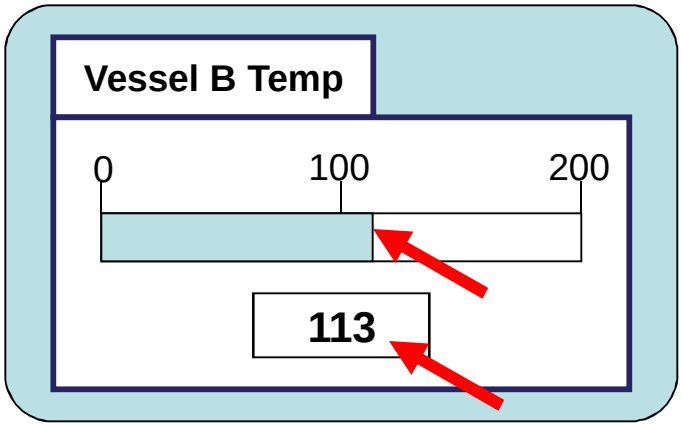
Context matters!

	office	industrial
type of data	textual	numeric
rate of change	slow	fast
environment	clean	dirty

... the oil soaked mouse!

Glass Interfaces ?

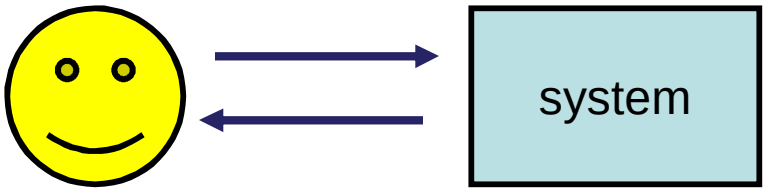
- industrial interface:
  - traditional … dials and knobs
  - now … screens and keypads
- glass interface
  - \+ cheaper, more flexible, multiple representations, precise values
  - – not physically located, loss of context, complex interfaces
- may need both



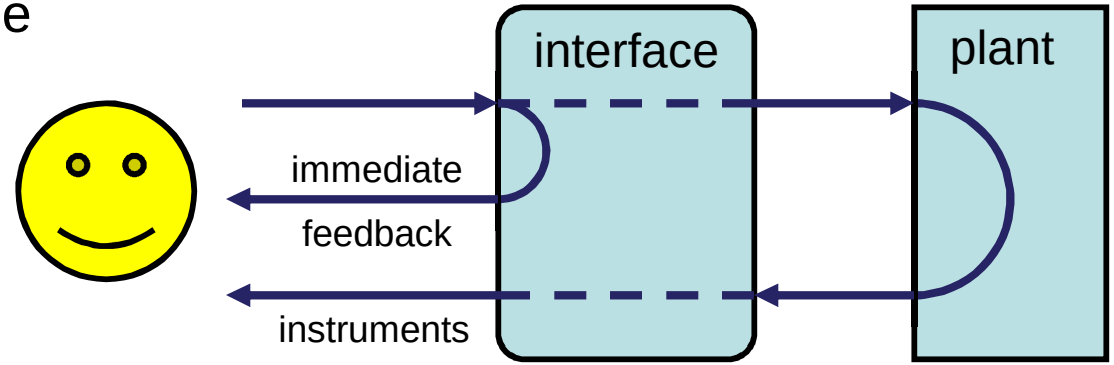
multiple representations of same information

Indirect Manipulation

- office– direct manipulation
  - user interacts with artificial world



- industrial – indirect manipulation
  - user interacts *with* real world *through* interface
- issues ..
  - feedback
  - delays



Interaction Styles

dialogue ... computer and user

distinct styles of interaction

Common Interaction Styles

- command line interface
- menus
- natural language
- question/answer and query dialogue
- form-fills and spreadsheets
- WIMP
- point and click
- three–dimensional interfaces

Command Line Interface

- Way of expressing instructions to the computer directly
  - function keys, single characters, short abbreviations, whole words, or a combination
- suitable for repetitive tasks
- better for expert users than novices
- offers direct access to system functionality
- command names/abbreviations should be meaningful!

Typical example: the Unix system

Menus

- Set of options displayed on the screen
- Options visible
  - less recall - easier to use
  - rely on recognition so names should be meaningful
- Selection by:
  - numbers, letters, arrow keys, mouse
  - combination (e.g. mouse plus accelerators)
- Often options hierarchically grouped
  - sensible grouping is needed
- Restricted form of full WIMP system

Natural Language

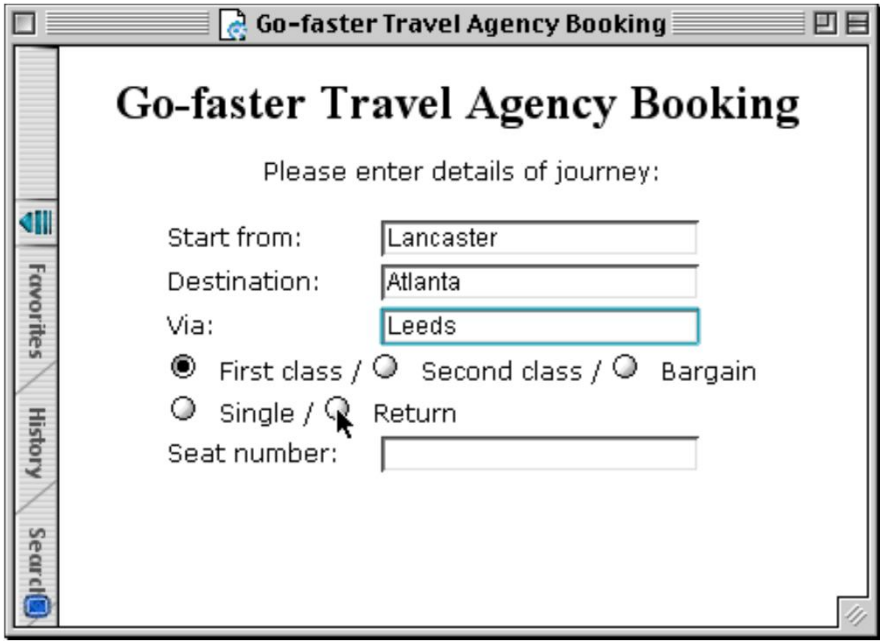
- Familiar to user
- speech recognition or typed natural language
- Problems
  - vague
  - ambiguous
  - hard to do well!
- Solutions
  - try to understand a subset
  - pick on key words

Query Interfaces

- Question/answer interfaces
  - user led through interaction via series of questions
  - suitable for novice users but restricted functionality
  - often used in information systems
- Query languages (e.g. SQL)
  - used to retrieve information from database
  - requires understanding of database structure and language syntax, hence requires some expertise

Form-fills

- Primarily for data entry or data retrieval
- Screen like paper form.
- Data put in relevant place
- Requires
  - good design
  - obvious correction facilities



Spreadsheets

- first spreadsheet VISICALC, followed by Lotus 1-2-3
  MS Excel most common today
- sophisticated variation of form-filling.
  - grid of cells contain a value or a formula
  - formula can involve values of other cells
    e.g. sum of all cells in this column
  - user can enter and alter data spreadsheet maintains consistency

WIMP Interface

Windows

Icons

Menus

Pointers

... or windows, icons, mice, and pull-down menus!

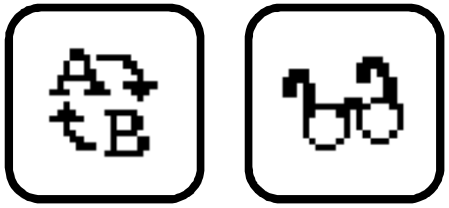
- default style for majority of interactive computer systems, especially PCs and desktop machines

Point and Click Interfaces

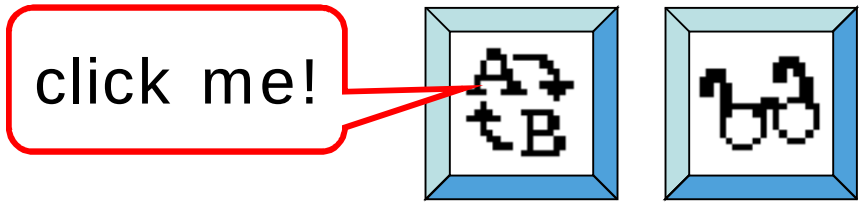
- used in ..
  - multimedia
  - web browsers
  - hypertext
- just click something!
  - icons, text links or location on map
- minimal typing

Three Dimensional Interfaces

- virtual reality
- ‘ordinary’ window systems
  - highlighting
  - visual affordance
  - indiscriminate use just confusing!
- 3D workspaces
  - use for extra virtual space
  - light and occlusion give depth
  - distance effects



flat buttons ...



... or sculptured

Elements of The Wimp Interface

windows, icons, menus, pointers

+ + +

buttons, toolbars,
palettes, dialog boxes

also see supplementary material
on choosing wimp elements

Windows

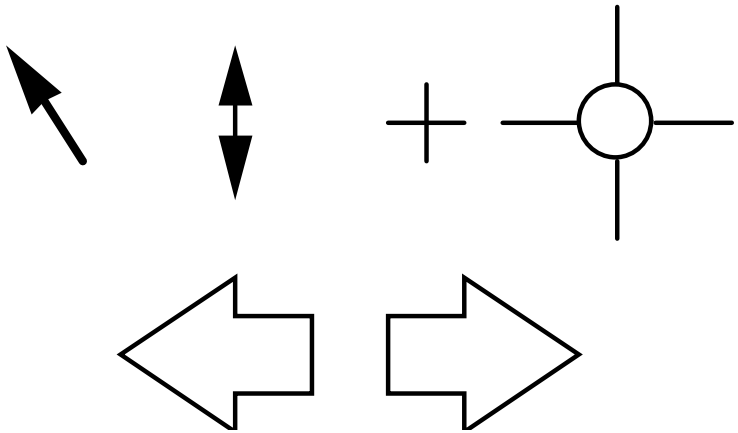
- Areas of the screen that behave as if they were independent
  - can contain text or graphics
  - can be moved or resized
  - can overlap and obscure each other, or can be laid out next to one another (tiled)
- scrollbars
  - allow the user to move the contents of the window up and down or from side to side
- title bars
  - describe the name of the window

Icons

- small picture or image
- represents some object in the interface
  - often a window or action
- windows can be closed down (iconised)
  - small representation fi many accessible windows
- icons can be many and various
  - highly stylized
  - realistic representations.

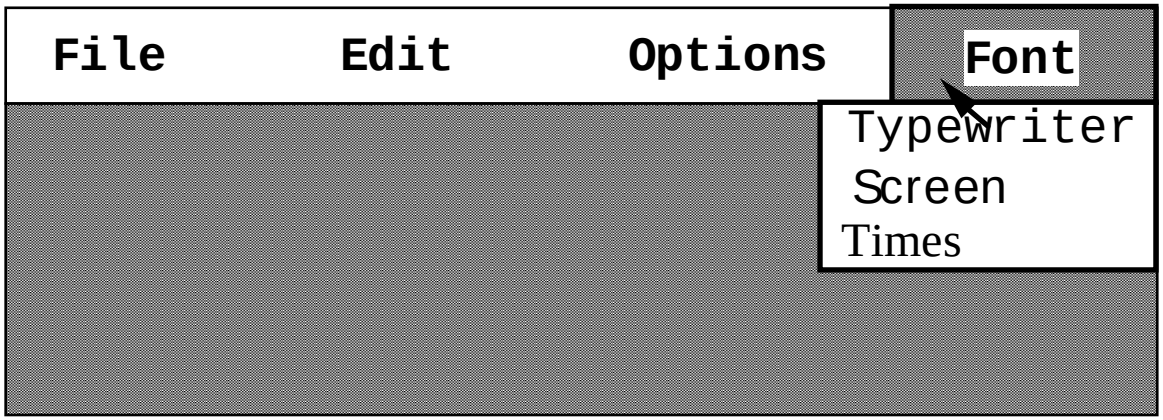
Pointers

- important component
  - WIMP style relies on pointing and selecting things
- uses mouse, trackpad, joystick, trackball, cursor keys or keyboard shortcuts
- wide variety of graphical images



Menus

- Choice of operations or services offered on the screen
- Required option selected with pointer



problem – take a lot of screen space

solution – pop-up: menu appears when needed

Kinds of Menus

- Menu Bar at top of screen (normally), menu drags down
  - pull-down menu - mouse hold and drag down menu
  - drop-down menu - mouse click reveals menu
  - fall-down menus - mouse just moves over bar!
- Contextual menu appears where you are
  - pop-up menus - actions for selected object
  - pie menus - arranged in a circle
    - easier to select item (larger target area)
    - quicker (same distance to any option)
      … but not widely used!

Menus extras

- Cascading menus
  - hierarchical menu structure
  - menu selection opens new menu
  - and so in ad infinitum
- Keyboard accelerators
  - key combinations - same effect as menu item
  - two kinds
    - active when menu open – usually first letter
    - active when menu closed – usually Ctrl + letter

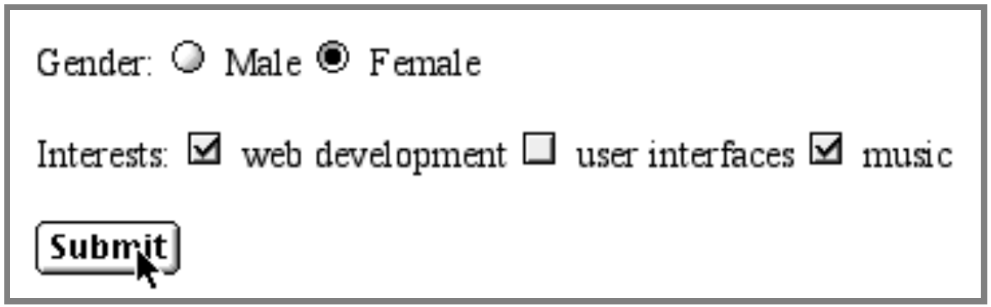
  usually different !!!

Menus Design Issues

- which kind to use
- what to include in menus at all
- words to use (action or description)
- how to group items
- choice of keyboard accelerators

Buttons

- individual and isolated regions within a display that can be selected to invoke an action



- Special kinds
  - radio buttons
    - set of mutually exclusive choices
  - check boxes
    - set of non-exclusive choices

Toolbars

- long lines of icons ...
  ... but what do they do?
- fast access to common actions
- often customizable:
  - choose *which* toolbars to see
  - choose *what* options are on it

Palettes and Tear-off Menus

- Problem
  - menu not there when you want it
- Solution
  - palettes – little windows of actions
    - shown/hidden via menu option
      e.g. available shapes in drawing package
  - tear-off and pin-up menus
    - menu ‘tears off’ to become palette

Dialogue Boxes

- information windows that pop up to inform of an important event or request information.

  e.g: when saving a file, a dialogue box is displayed to allow the user to specify the filename and location. Once the file is saved, the box disappears.

Interactivity

easy to focus on look

what about feel?

Speech–driven Interfaces

- rapidly improving …
  … but still inaccurate
- how to have robust dialogue?
  … interaction of course!

e.g. airline reservation:
reliable “yes” and “no”
+ system reflects back its understanding
“you want a ticket from New York to Boston?”

Look and … Feel

- WIMP systems have the same elements:

  windows, icons., menus, pointers, buttons, etc.

- but different window systems

  … *behave* differently

  e.g. MacOS vs Windows menus

appearance + behaviour = look and feel

Error and Repair

can't always avoid errors …
… but we can put them right

make it easy to *detect* errors
… then the user can *repair* them